Cisco Project 1 the basics

Introduction

This is a part of a networking project series that I will be working on to further recap the knowledge I gained in the last 3 years as a computer science student. Each project will have either a readme file or a pdf like this that will document each project and the walkthrough behind it.

Physical Topology Setup

Commit: "complete physical topology setup with switches routers and pcs"
In this phase, I set up the physical topology of a simple network between 4 PCs using four routers (Router-PT) and four switches (Switch-PT / 2960-24TT), all linked together with automatic cable identification provided by Cisco Packet Tracer.

Logical Topology & Addressing Scheme

The network is partitioned into four distinct LAN subnets and two point-to-point WAN subnets:

- **PC0:** 192.168.1.1/24 (Gateway: 192.168.1.2)
- **PC1:** 192.168.1.3/24 (Gateway: 192.168.1.4)
- **PC2:** 192.168.2.1/24 (Gateway: 192.168.2.2)
- **PC3:** 192.168.3.1/24 (Gateway: 192.168.3.2)
- **WAN Subnet 1 (Top Link):** 10.0.0.0/30 (Router0 10.0.0.1 & Router2 10.0.0.2)
- **WAN Subnet 2 (Bottom Link):** 10.0.1.0/30 (Router1 10.0.1.1 & Router3 10.0.1.2)

Side Note on Subnet Masks:

- LAN interfaces use a 255.255.255.0 mask (/24), allowing up to 254 usable host IP addresses.
- Point-to-point router WAN links use a 255.255.255.252 mask (/30) to allocate exactly 2 usable IP addresses per link alongside the standard network and broadcast addresses.
- PC0 and PC1 reside in the same subnet (192.168.1.0/24) connected via two interconnected switches, serving as an inter-switch bridge between Router0 and Router1.

Interface Configuration via Cisco CLI

1. Router0 Configuration

```
Router> enable
Router# config terminal
Router(config)# interface FastEthernet0/0
Router(config-if)# ip address 192.168.1.2 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface Serial2/0
```

```
Router(config-if)# ip address 10.0.0.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)# end
Router# write memory
```

2. Router1 Configuration

```
Router> enable
Router# config terminal
Router(config)# interface FastEthernet0/0
Router(config-if)# ip address 192.168.1.4 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface Serial2/0
Router(config-if)# ip address 10.0.1.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)# end
Router# write memory
```

Static Routing Configuration

To establish full inter-subnet reachability across the entire topology, static routes (ip route) are explicitly configured on every router. Each static route specifies the target destination network, the destination subnet mask, and the immediate next-hop IP address of the adjacent router interface.

1. Router0 Static Routes

Router0 forwards traffic destined for LAN 2 across the top WAN link, and routes traffic for LAN 3 and the bottom WAN link through Router1 across the switch trunk:

```
Router0# config terminal
Router0(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router0(config)# ip route 192.168.3.0 255.255.255.0 192.168.1.4
Router0(config)# ip route 10.0.1.0 255.255.255.252 192.168.1.4
Router0(config)# end
Router0# write memory
```

2. Router1 Static Routes

Router1 forwards traffic destined for LAN 3 across the bottom WAN link, and routes traffic for LAN 2 and the top WAN link through Router0 across the switch trunk:

```
Router1# config terminal
Router1(config)# ip route 192.168.3.0 255.255.255.0 10.0.1.2
Router1(config)# ip route 192.168.2.0 255.255.255.0 192.168.1.2
Router1(config)# ip route 10.0.0.0 255.255.255.252 192.168.1.2
Router1(config)# end
```

```
Router1# write memory
```

3. Router2 Static Routes

Router2 routes all return traffic destined for LAN 1, LAN 3, and the bottom WAN link back to Router0 across the top WAN link:

```
Router2# config terminal
Router2(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router2(config)# ip route 192.168.3.0 255.255.255.0 10.0.0.1
Router2(config)# ip route 10.0.1.0 255.255.255.252 10.0.0.1
Router2(config)# end
Router2# write memory
```

4. Router3 Static Routes

Router3 routes all return traffic destined for LAN 1, LAN 2, and the top WAN link back to Router1 across the bottom WAN link:

```
Router3# config terminal
Router3(config)# ip route 192.168.1.0 255.255.255.0 10.0.1.1
Router3(config)# ip route 192.168.2.0 255.255.255.0 10.0.1.1
Router3(config)# ip route 10.0.0.0 255.255.255.252 10.0.1.1
Router3(config)# end
Router3# write memory
```

Routing Table & Path Summary

Router	Target Network	Subnet Mask	Next-Hop IP	Description
Router0	192.168.2.0	255.255.255.0	10.0.0.2	Direct top WAN path to PC2 LAN
Router0	192.168.3.0	255.255.255.0	192.168.1.4	Inter-switch path to Router1 for PC3 LAN
Router1	192.168.3.0	255.255.255.0	10.0.1.2	Direct bottom WAN path to PC3 LAN
Router1	192.168.2.0	255.255.255.0	192.168.1.2	Inter-switch path to Router0 for PC2 LAN
Router2	192.168.1.0	255.255.255.0	10.0.0.1	Return WAN path to Router0 for LAN 1
Router3	192.168.1.0	255.255.255.0	10.0.1.1	Return WAN path to Router1 for LAN 1

Verification & Results

End-to-end connectivity was verified using Cisco Packet Tracer ICMP Simulation and Command Prompt ping tests:

- PC0 → PC2 (192.168.2.1): **Successful** (0% packet loss)
- PC0 → PC3 (192.168.3.1): **Successful** (0% packet loss)
- PC1 → PC2 (192.168.2.1): **Successful** (0% packet loss)
- PC1 → PC3 (192.168.3.1): **Successful** (0% packet loss)

All four subnets demonstrate full inter-subnet reachability across both WAN serial links and the central inter-switch trunk.